



Open Science Autonomy Lab

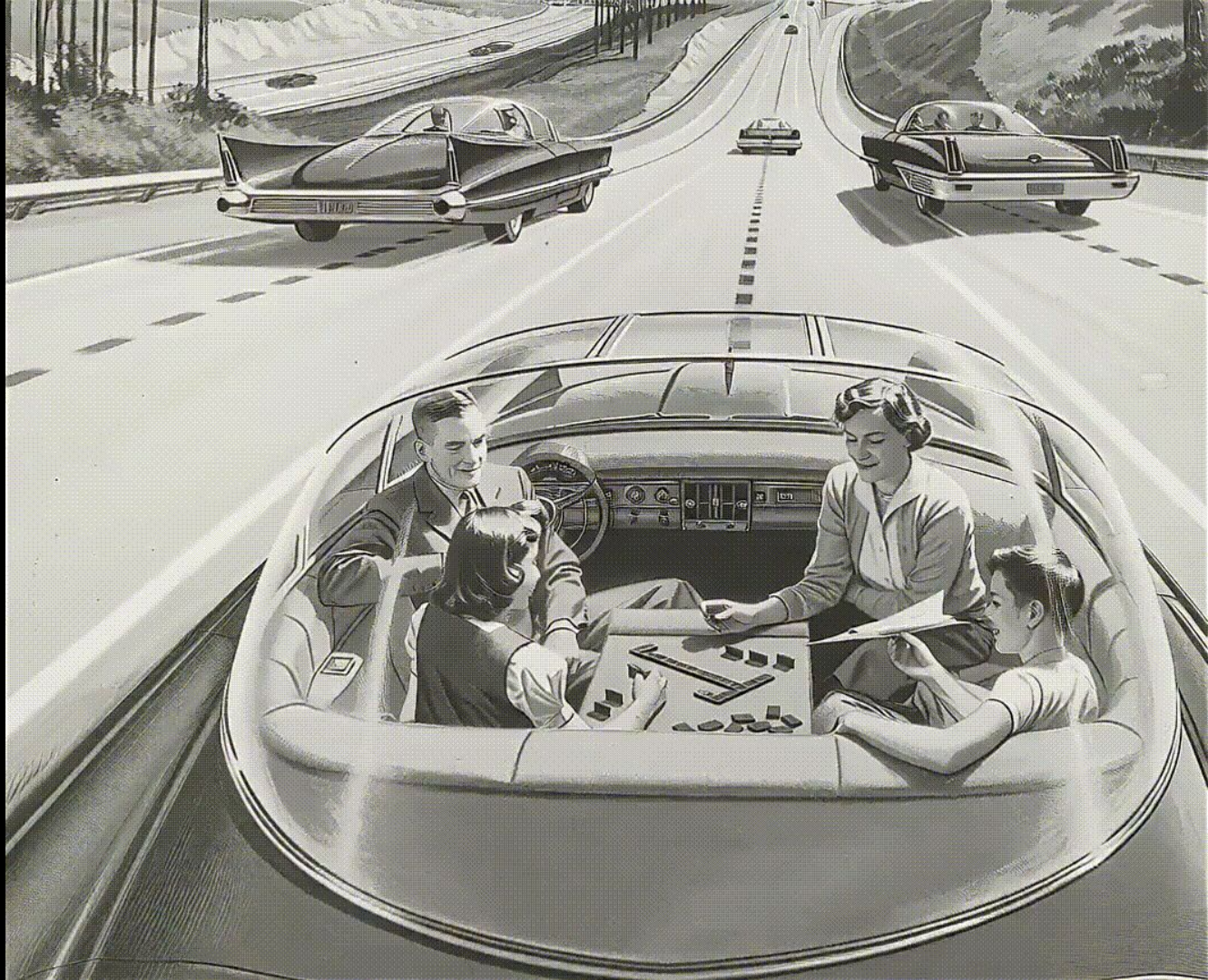
Kyutai ELLIS

Scalable Autonomous Intelligence

<https://kesai.eu>

CONFIDENTIAL

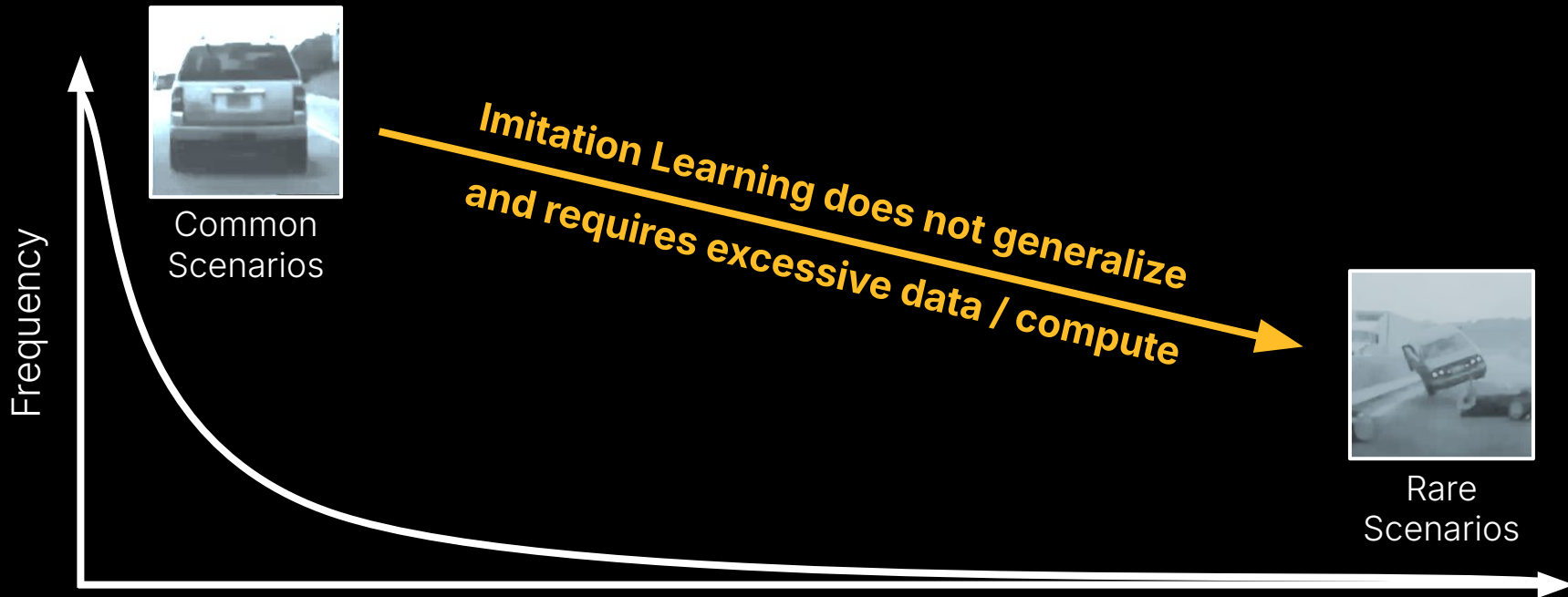
**Driverless Car
of the Future**
Saturday Evening
Post, 1956



Where are we today?

**The automotive industry spent
>\$100B in the last 10 years to
automate 0.003% of traffic***

Limits of Imitation Learning



The Problem

Robot learning is bottlenecked by the cost of physical interaction

- **Imitation learning** requires excessive amount of expert data and compute
- Humans are orders of magnitude more **sample efficient** than today's AI
- Robotics is dominated by **proprietary silos** with closed data and code

Our Mission

Establish KE:SAI as a joint Franco-German non-profit open science lab

- Pioneer world-leading research in **causal world models** and **autonomy**
- Develop first **fully open data- and compute-efficient self-driving stack**
- Expansion to **other areas of robotics**

Fully Open-Source Autonomy

- Open weights, data, code, simulators & **detailed, impactful papers**
- **Reproducible** with moderate human and compute resource requirements

Frontier Research on Causal World Models

- **Efficient, hierarchical & latent** world models that are physically grounded
- Identification of **causal mechanisms** and application of **interventions**

Deployable Sim2Real Reinforcement Learning

- **Highly parallelized** policy learning from heterogeneous data (sim/real)
- **Foundation policies** operational across **embodiments** and **environments**

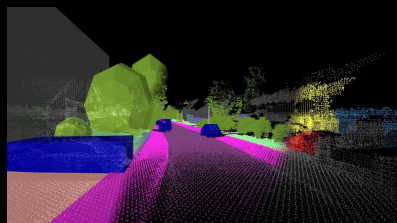
Towards Efficient Physical AI – Our Expertise



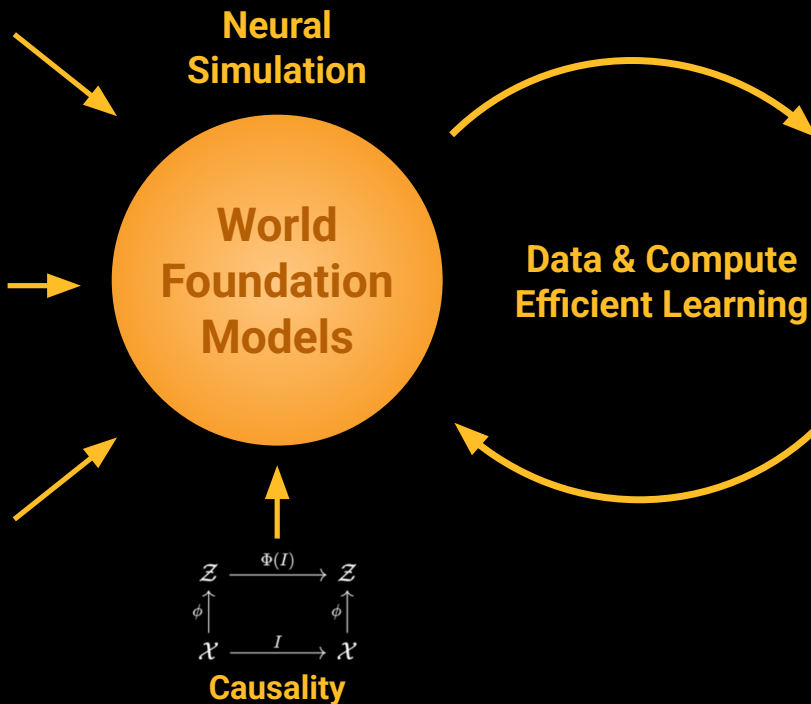
3D Computer Vision



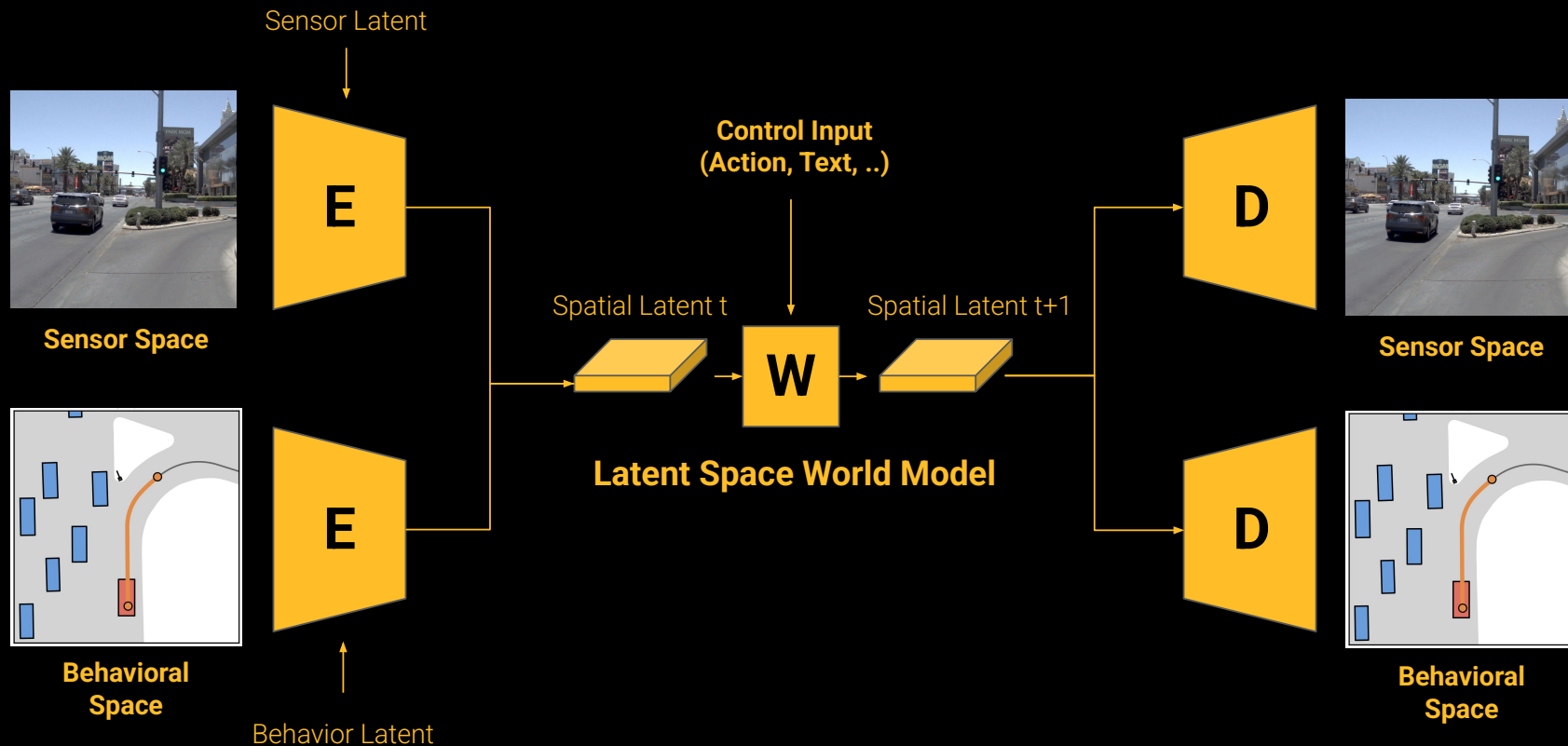
Generative Models



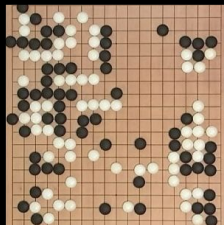
Datasets & Benchmarks



Towards Efficient Physical AI – Latent World Models



Towards Efficient Physical AI – The Rise of RL



Go

DeepMind, 2017



Drones

Univ. Zürich, 2023

- ▶ Robust driving
- ▶ No Expert
- ▶ Efficiency
- ▶ Generalization



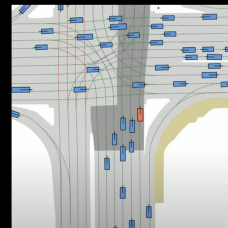
Atari

DeepMind, 2015



Quadrupeds

ETHZ, 2020



Self-Driving

Univ. Tübingen, 2025

Apple, 2025

← RL
← Self-Play

Towards Efficient Physical AI – Synthetic Data



Ground-truth



Controlled by non-expert action

The KE:SAI Lab



- Joint venture of **kyutai Paris** and **ELLIS Institute Tübingen**
- **Co-located** in Tübingen (headquarters) and Paris
- Merging kyutai's **open science** ethos and **foundation model** expertise with ELLIS' **talent attraction** and **foundational research excellence** in 3D computer vision, data-driven simulation, causality and physical AI
- **Strategic partnerships** with kyutai, NVIDIA AV Research, General Intuition, ...

Founding Team



Andreas Geiger

Chief Executive Officer
Co-Founder

[ex-MPI, ETHZ, UT]

Expertise

3D Vision, Self-Driving,
Generative Models



Kashyap Chitta

Chief Technology Officer
Co-Founder

[ex-CMU, NVIDIA, UT]

Expertise

Robotics, Self-Driving,
World Models



Bernhard Schölkopf

Chief Causality Officer
Co-Founder

[MPI, ETHZ, ELLIS]

Expertise

Machine Learning,
Causality



Bernhard Jaeger

Technical Staff
Co-Founder

[ex-TUM, UT]

Expertise

Self-Driving,
Simulation



Daniel Dauner

Technical Staff
Co-Founder

[ex-UT]

Expertise

Self-Driving,
Benchmarking

Track Record

- **27 best paper awards** at leading vision, learning and robotics conferences
- Introduction of **KITTI benchmarks** which transformed vision & self-driving
- Top places at **9 self-driving competitions** (nuPlan, CARLA, Waymo, ...)
- Open-sourced 2 largest (1700h+) **driving datasets** (OpenDV, PhysicalAI-AV)
- First **reinforcement learning** planner with state-of-the-art driving (CaRL)
- First open large-scale **video world model** for driving (Vista)
- First **generative simulator** for vectorized driving environments (SLEDGE)
- First **pseudo-simulation** benchmark for autonomous driving (NAVSIM)
- Pioneered methods for **causal discovery** from observational data
- Work on **implicit models** triggered paradigm shift in vision and graphics
- Creators of **ELLIS** and the **ELLIS PhD & PostDoc program**
- **Attracting talent:** former lab members co-founded successful startups
Black Forest Labs, SpAltial AI, Mirelo AI, ...

Funding

Founding donations: € 25M

- Elite team, no academic salary caps
- Excellent compute infrastructure (300+ GPUs)
- Research platform for real-world deployment
- 5y runway

Additional resources:

- New donations (Goal: € 100M within 2 years)
- Commercial and public partnerships

Our Founding Donors



Eric Schmidt



Xavier Niel



Rodolphe Saadé



Team Composition

Leadership: (3)

- Andreas Geiger (CEO)
- Kashyap Chitta (CTO)
- Bernhard Schölkopf* (CCO)

Technical Staff: (8)

- Bernhard Jaeger (co-founder)
- Daniel Dauner (co-founder)
- Haofei Xu
- 5x N.N. (Paris/Tübingen)

Junior Staff: (2)

- 2x N.N. (Paris/Tübingen)

PhD Students: (5)

- Long Nguyen*
- 4x N.N. (Tübingen)

Admin: (1)

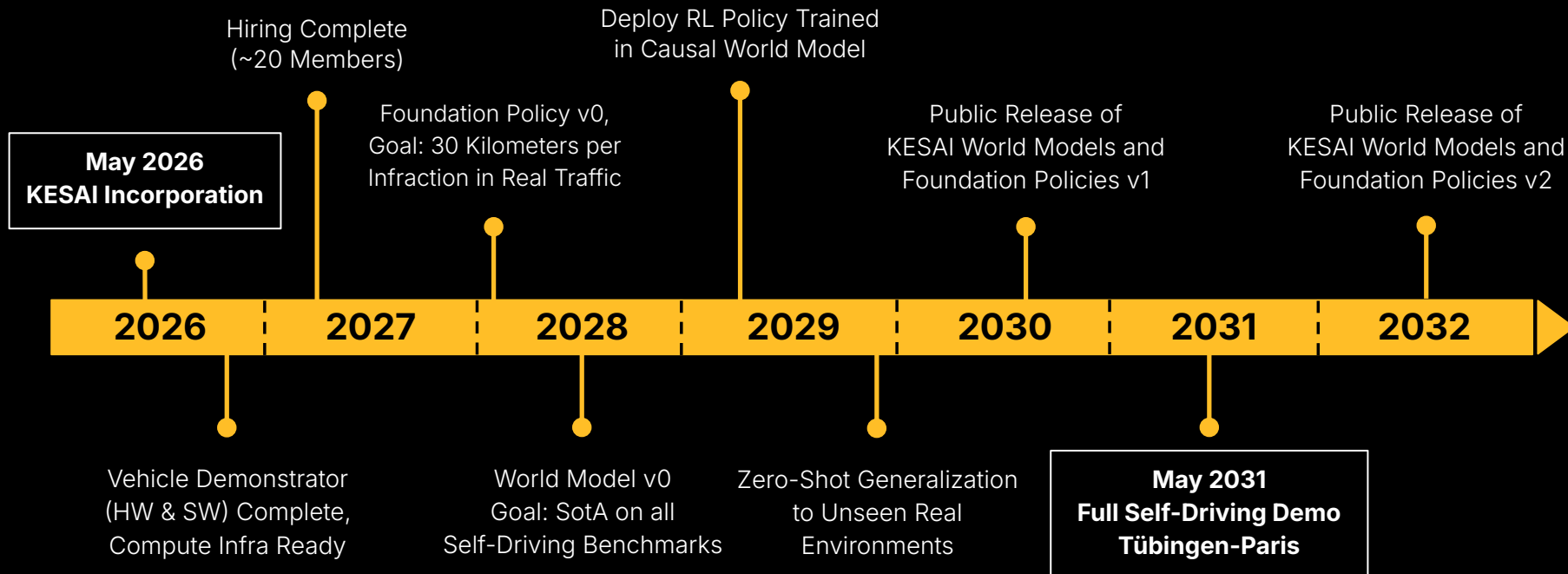
- N.N. (Tübingen)

Headquarters



Technology Park Tübingen – Maria-von-Linden-Str. 1 – 72076 Tübingen – Germany

Timeline



Adopting Kyutai's Fluid Model

